EVERY REMAINDER EVERYWHERE: A DIRECT PROOF ON THE ATOMIC INVERSE TREE

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ABSTRACT. We prove again that for every prime p, every positive integer w not divisible by 3, every residue $r \in \mathbb{Z}/p\mathbb{Z}$ and every bound B, some $n > B$ reaches w under the Collatz map with $n \equiv r \pmod p$. This time we work with the two *atomic* backward steps only, $n \mapsto 2n$ and $n \mapsto (n-1)/3$, with no change of coordinates and no left/right multiplier tables. States are pairs (residue modulo 3^{K+1}, parity). Stabilization of fibre sizes is proved as before. The step that replaces the case analysis on dilations is a clean obstruction: no assignment of residues modulo p to states can satisfy all edge relations (a "flat assignment"), proved by a short argument with powers of 2 and 4 and the lifting-the-exponent lemma. The obstruction forces the affine holonomy group to contain a non-trivial translation, and translations fill every fibre. The mathematical argument is written to follow the accompanying Lean 4 file `AtomicERE.lean` step by step.

CONVENTIONS

p is a prime. From Section 4 to Section 8 we assume $p \geq 5$, so that 2 and 3 are invertible in $\mathbb{Z}/p\mathbb{Z}$. For $n \in \mathbb{N}$, $(n : \mathbb{Z}/p\mathbb{Z})$ is its residue. All divisions of naturals below are exact. We write $M_K = 3^{K+1}$. A residue is a *unit* if it is not divisible by 3. Every statement in this paper has a counterpart in `AtomicERE.lean`; names are listed in the appendix. In several places the Lean file is more economical than the earlier coordinate-based paper, and we follow the Lean file.

1. THE COLLATZ MAP AND THE ATOMIC INVERSE STEPS

Let $\mathcal{C}(n) = n/2$ for even n and $\mathcal{C}(n) = 3n+1$ for odd n. Write $n \rightsquigarrow v$ if $\mathcal{C}^{(k)}(n) = v$ for some $k \geq 0$.

Theorem 1.1 (Main theorem)**.** *Let p be prime, $w > 0$ with $3 \nmid w$, $r \in \mathbb{Z}/p\mathbb{Z}$ and $B \in \mathbb{N}$. Then there is $n > B$ with $n \rightsquigarrow w$ and $(n : \mathbb{Z}/p\mathbb{Z}) = r$.*

Lemma 1.2 (Atomic inverse steps)**.**
(1) $\mathcal{C}(2a) = a$ and $\mathcal{C}^{(j)}(2^j m) = m$.
(2) $\rightsquigarrow$ is transitive.
(3) $\mathcal{C}(m) = n$ iff $m = 2n$, or m is odd and $3m+1 = n$.

Proof. Immediate from the definition of $\mathcal{C}$ read backwards. □

Corollary 1.3. *If $3 \mid n$ and $\mathcal{C}(m) = n$ then $m = 2n$.*

Proof. If m is odd with $3m+1 = n$ then $n \equiv 1 \pmod 3$. □

So a number has two possible predecessors: the *double* $2n$, always, and the *undo* $(n-1)/3$, exactly when $n \equiv 4 \pmod 6$ (then $n-1 = 6k+3$ and $(n-1)/3 = 2k+1$ is odd). Numbers divisible by 3 admit only doubling, which is why $3 \nmid w$ is assumed. We call a node *live* if $3 \nmid n$. The undo step is only used when it produces a live node; this is the condition $(n-1)/3 \not\equiv 0 \pmod 3$, i.e. $n \not\equiv 1 \pmod 9$.

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2. The atomic tree

Definition 2.1 (Atomic tree)**.** For $v \in \mathbb{N}$, $\mathcal{T}(v) \subseteq \mathbb{N}$ is the smallest set with

- $v \in \mathcal{T}(v)$;
- if $n \in \mathcal{T}(v)$ then $2n \in \mathcal{T}(v)$ (step D);
- if $n \in \mathcal{T}(v)$, $n \equiv 4 \pmod 6$ and $\frac{n-1}{3} \not\equiv 0 \pmod 3$, then $\frac{n-1}{3} \in \mathcal{T}(v)$ (step U).

Lemma 2.2. *Let $v \in \mathbb{N}$.*

(1) If $3 \nmid v$ then $3 \nmid n$ for all $n \in \mathcal{T}(v)$.

(2) Every $n \in \mathcal{T}(v)$ satisfies $n \rightsquigarrow v$.

(3) If v is even and $n \in \mathcal{T}(v)$ is odd, then $n = (m-1)/3$ for some $m \in \mathcal{T}(v)$ with $m \equiv 4 \pmod 6$ and $(m-1)/3 \not\equiv 0 \pmod 3$.

Proof. (1) By induction: doubling preserves liveness, and U is only allowed into live numbers. (2) By induction using Lemma 1.2: $2n \rightsquigarrow n$ in one step, and $\mathcal{C}((n-1)/3) = 3 \cdot \frac{n-1}{3} + 1 = n$ because $(n-1)/3$ is odd. (3) The seed v is even and D produces even numbers, so an odd element must come from U. □

For $p \geq 5$ we use the seed $v = 2w$, which is even and live; $2w \rightsquigarrow w$ by Lemma 1.2(1). There is no seed table.

3. Arithmetic modulo 3^{K+1}

Lemma 3.1 (Lifting the exponent for base 4)**.** *For $m \geq 1$, $v_3(4^m - 1) = 1 + v_3(m)$. Hence for all K, m:*

$$4^m \equiv 1 \pmod{3^{K+1}} \iff 3^K \mid m.$$

Proof. This is the classical lemma for $(4, 1)$ at the prime 3. For $m \geq 1$: $4^m \equiv 1 \pmod{3^{K+1}}$ iff $1 + v_3(m) \geq K + 1$. □

Lemma 3.2 (2 generates the units)**.**

(1) $2^m \equiv 1 \pmod{3^{K+1}}$ iff $2 \cdot 3^K \mid m$.

(2) If $E_1, E_2 < 2 \cdot 3^K$ and $2^{E_1} \equiv 2^{E_2} \pmod{3^{K+1}}$ then $E_1 = E_2$.

(3) For units $c, d < 3^{K+1}$ there is j with $2^j c \equiv d \pmod{3^{K+1}}$.

Proof. (1) If $2^m \equiv 1$, reducing mod 3 forces m even (odd powers of 2 are $\equiv 2$), so $m = 2n$ and $4^n \equiv 1$; Lemma 3.1 gives $3^K \mid n$. The converse is Lemma 3.1 again. (2) Cancel the smaller power (coprime to 3) and apply (1). (3) The map $E \mapsto 2^E c \bmod 3^{K+1}$ on $\{0, \ldots, 2 \cdot 3^K - 1\}$ lands in the units and is injective by (2). There are exactly $2 \cdot 3^K$ units below 3^{K+1}, so it is onto. □

Lemma 3.3 (A congruence for $4^{3^{K-1}}$)**.** *For $K \geq 1$: $4^{3^{K-1}} \equiv 1 + 3^K \pmod{3^{K+1}}$.*

Proof. We show $4^{3^n} = 1 + 3^{n+1} + 3^{n+2}q$ for some q, by induction. For $n = 0$, $4 = 1 + 3$. If $4^{3^n} = 1 + X$ with $X = 3^{n+1}(1 + 3q)$, then $(1 + X)^3 = 1 + 3X + 3X^2 + X^3$. Here $3X = 3^{n+2} + 3^{n+3}q(\ldots)$ contributes 3^{n+2} plus a multiple of 3^{n+3}, while $3X^2$ and X^3 are multiples of 3^{n+3}. Take $n = K - 1$. □

This is the linear-term statement of the binomial expansion; the quotient $(4^{3^{K-1}} - 1)/3^K$ is $\equiv 1 \pmod 3$, not merely a 3-adic unit.

Lemma 3.4 (Units that are $\equiv 1 \pmod 3$ are powers of 4)**.** *If $c < 3^{K+1}$ and $c \equiv 1 \pmod 3$ then $c \equiv 4^e \pmod{3^{K+1}}$ for some e.*

Proof. By Lemma 3.2(3), $c \equiv 2^j$. If j were odd, $2^j \equiv 2 \pmod 3$. So $j = 2e$. □

4. States, fibres and stabilization

Fix an even seed v with $3 \nmid v$ and $p \geq 5$.

Definition 4.1 (Fibre)**.** For $K \geq 1$, a unit c and $f \in \{0, 1\}$,

$$F_K(c, f) = \{(n : \mathbb{Z}/p\mathbb{Z}) \ : \ n \in \mathcal{T}(v),\ n \equiv c \pmod{3^{K+1}},\ n \equiv f \pmod 2\}.$$

The pair (c, f) is a *state*: c is a unit residue modulo 3^{K+1} and f a parity flag.

Lemma 4.2 (Elementary fibre facts)**.**

(1) (Projection) For $J \leq K$, $F_K(c, f) \subseteq F_J(c \bmod 3^{J+1}, f)$.

(2) (Doubling) $t \in F_K(c, f) \Rightarrow 2t \in F_K(2c \bmod 3^{K+1}, 0)$. Hence $|F_K(c, f)| \leq |F_K(2c, 0)|$, since $t \mapsto 2t$ is injective.

(3) (Equal sizes) For units c, d: $|F_K(c, 0)| = |F_K(d, 0)|$.

(4) (Exact doubling) $F_K(2c, 0) = 2 \cdot F_K(c, 0)$.

(5) (Odd inside even) $F_K(c, 1) \subseteq F_K(c, 0)$.

(6) (Undo) For a unit $t_0 < 3^{K+1}$ and $t \in \mathbb{Z}/p\mathbb{Z}$:

$$t \in F_K(t_0, 1) \iff 3t + 1 \in F_{K+1}((3t_0 + 1) \bmod 3^{K+2}, 0).$$

Proof. (1) $3^{J+1} \mid 3^{K+1}$. (2) If $n \in \mathcal{T}(v)$ realizes t, then $2n \in \mathcal{T}(v)$ realizes $2t$ in class $2c$ with flag 0. (3) By Lemma 3.2(3), $d \equiv 2^j c$ and $c \equiv 2^i d$; iterating (2) gives $|F_K(c, 0)| \leq |F_K(d, 0)| \leq |F_K(c, 0)|$. (4) By (2) the image $2 \cdot F_K(c, 0)$ lies in $F_K(2c, 0)$ and the two sets have the same size by (3). (5) If $t \in F_K(c, 1)$, then $2t \in F_K(2c, 0) = 2 \cdot F_K(c, 0)$ by (2) and (4); cancel 2. (6) For ($\Rightarrow$): an odd $n \in \mathcal{T}(v)$ in class t_0 equals $(m - 1)/3$ with $m = 3n + 1 \in \mathcal{T}(v)$ by Lemma 2.2(3), and $m \equiv 3t_0 + 1 \pmod{3^{K+2}}$ because $n \equiv t_0 \pmod{3^{K+1}}$; also $(m : \mathbb{Z}/p\mathbb{Z}) = 3t + 1$. For ($\Leftarrow$): let $m \in \mathcal{T}(v)$ be even with $m \equiv 3t_0 + 1 \pmod{3^{K+2}}$ and $(m : \mathbb{Z}/p\mathbb{Z}) = 3t + 1$. Then $m \equiv 1 \pmod 3$ and $m \equiv 3t_0 + 1 \not\equiv 1 \pmod 9$ because t_0 is a unit, so $m \equiv 4 \pmod 6$ and U applies: $n = (m - 1)/3 \in \mathcal{T}(v)$ is odd. From $m = 3n + 1$ and $m \equiv 3t_0 + 1 \pmod{3^{K+2}}$ we get $n \equiv t_0 \pmod{3^{K+1}}$, and $3(n : \mathbb{Z}/p\mathbb{Z}) + 1 = 3t + 1$ gives $(n : \mathbb{Z}/p\mathbb{Z}) = t$ since 3 is invertible. □

Definition 4.3. $\sigma_K = |F_K(1, 0)|$.

Lemma 4.4.

(1) $\sigma_{K+1} \leq \sigma_K$ and $\sigma_K \geq 1$.

(2) For a unit $t_0 < 3^{K+1}$: $|F_K(t_0, 1)| = \sigma_{K+1}$.

Proof. (1) Monotonicity is projection (Lemma 4.2(1)) at class 1. Non-emptiness: $v \in \mathcal{T}(v)$ lies in some even fibre, and Lemma 4.2(3) transports this to class 1. (2) By Lemma 4.2(6), $F_K(t_0, 1)$ is the preimage of $F_{K+1}(e, 0)$ under the injection $t \mapsto 3t + 1$, and that fibre is contained in the range of this map; so its size is $|F_{K+1}(e, 0)| = \sigma_{K+1}$ by (3). □

Proposition 4.5 (Stabilization)**.** *There is $K_0 \geq 1$ such that for all $K \geq K_0$:*

(a) $F_K(c, 1) = F_K(c, 0)$ for every unit $c < 3^{K+1}$;

(b) $F_{K+1}(e, 0) = F_K(e \bmod 3^{K+1}, 0)$ for every unit $e < 3^{K+2}$.

Proof. A non-increasing sequence of positive integers is eventually constant: choose $K_0 \geq 1$ with $\sigma_K = \sigma_{K_0}$ for $K \geq K_0$. (a) $F_K(c, 1) \subseteq F_K(c, 0)$ by Lemma 4.2(5); their sizes are σ_{K+1} and σ_K by Lemma 4.4(2) and Lemma 4.2(3), equal for $K \geq K_0$. (b) $F_{K+1}(e, 0) \subseteq F_K(e \bmod 3^{K+1}, 0)$ by projection; both have size $\sigma_{K+1} = \sigma_K$. □

5. Affine labels and holonomy

$\mathrm{Aff}(p) = \mathbb{Z}/p\mathbb{Z}^\times \times \mathbb{Z}/p\mathbb{Z}$ acts by $f(t) = f_1 t + f_2$, with $f\,;g = (g_1 f_1,\ g_1 f_2 + g_2)$ so that $(f\,;g)(t) = g(f(t))$, identity $(1,0)$, and $f^{-1} = (f_1^{-1}, -f_1^{-1} f_2)$.

Definition 5.1 (Edges, walks, holonomy)**.** Fix $K \geq 1$. The vertices are the states (c, f). There are two kinds of edges.

- **Double edges.** For a unit $c < 3^{K+1}$ and $f \in \{0, 1\}$: $(c, f) \to (2c \bmod 3^{K+1}, 0)$ with label $(2, 0)$.
- **Undo edges.** For $e < 3^{K+2}$ with $e \equiv 1 \pmod 3$ and $e \not\equiv 1 \pmod 9$: $(e \bmod 3^{K+1}, 0) \to (\frac{e-1}{3} \bmod 3^{K+1}, 1)$ with label $(3^{-1}, -3^{-1})$.

A *walk* is a finite sequence of edges traversed forward or backward; its label is the composite of the labels (inverse label for a backward traversal). Let $\mathrm{Hol}(K)$ be the set of labels of closed walks at base $= (1, 0)$. It is a subgroup of $\mathrm{Aff}(p)$. Every label has non-zero slope.

The label of an undo edge is exactly the inverse of $t \mapsto 3t + 1$, matching Lemma 4.2(6).

Proposition 5.2 (Exact label action)**.** *Let $K \geq K_0$. For every edge $u \to w$ with label f: $f(F_K(u,)) = F_K(w,)$. Consequently, for every walk from u to w with label g, $g(F_K(u,)) = F_K(w,)$; in particular each $h \in \mathrm{Hol}(K)$ maps $F_K(1, 0)$ onto itself.*

Proof. Double edge from (c, f_0): by Proposition 4.5(a) $F_K(c, f_0) = F_K(c, 0)$, and $2 \cdot F_K(c, 0) = F_K(2c, 0)$ by Lemma 4.2(4). Undo edge from e: by Proposition 4.5(b), $F_K(e \bmod 3^{K+1}, 0) = F_{K+1}(e, 0)$, and by Lemma 4.2(6) with $t_0 = \frac{e-1}{3}$ (a unit below 3^{K+1} since $e \not\equiv 1 \pmod 9$) this is carried by $t \mapsto 3^{-1}t - 3^{-1}$ onto $F_K(t_0, 1)$. For walks, induct on length, using that backward traversals invert a bijection. □

Lemma 5.3 (Connectedness)**.** *For $K \geq 1$, every state (c, f) is joined to base by a walk.*

Proof. Even states: by Lemma 3.2(3) there is j with $2^j \equiv c$; follow j double edges from base. Odd states $(c, 1)$: take $e = 3c + 1 < 3^{K+2}$; then $e \equiv 1 \pmod 3$, $e \not\equiv 1 \pmod 9$ since $3 \nmid c$, and the undo edge from $(e \bmod 3^{K+1}, 0)$ ends at $(c, 1)$. □

6. No flat assignment

This section is the replacement for the case analysis on dilation orders in the coordinate-based proof. It uses no holonomy, only the edge relations.

Definition 6.1 (Flat assignment)**.** A function Φ from states to $\mathbb{Z}/p\mathbb{Z}$ is *flat* at depth K if $\Phi(w) = f(\Phi(u))$ for every edge $u \to w$ with label f. Explicitly,

$$\Phi(2c \bmod M_K, 0) = 2\Phi(c, f), \qquad 3\,\Phi(\tfrac{e-1}{3} \bmod M_K, 1) = \Phi(e \bmod M_K, 0) - 1. \tag{6.1}$$

Theorem 6.2 (No flat assignment)**.** *Let $p \geq 5$ and $K \geq 1$. There is no flat assignment at depth K.*

The proof is in six steps. Suppose Φ is flat and write $\phi(c) = \Phi(c, 0)$.

Step 1 (collapse and normalization). Applying the double relation at $f = 0$ and $f = 1$ gives $2\Phi(c, 0) = 2\Phi(c, 1)$, so $\Phi(c, 0) = \Phi(c, 1)$ for all units c. Then $\phi(2c) = 2\phi(c)$. The undo relation with $e = 4$ (valid: $4 \equiv 1 \pmod 3$, $4 \not\equiv 1 \pmod 9$, $4 < 3^{K+2}$) gives $3\Phi(1, 1) = \phi(4) - 1$. With $\phi(4) = 4\phi(1)$ and $\Phi(1, 1) = \phi(1)$ we get $3\phi(1) = 4\phi(1) - 1$, so $\phi(1) = 1$.

Step 2 (powers of 2). By induction, $\phi(2^j \bmod M_K) = 2^j$ in $\mathbb{Z}/p\mathbb{Z}$ for all $j \geq 0$. There is nothing to check about consistency around the cycle: the left side depends only on $2^j \bmod M_K$,

and the relation is only ever used for the representatives 2^j that occur. (In particular, applying it at $j = 2 \cdot 3^K$ shows $2^{2\cdot 3^K} = 1$ in $\mathbb{Z}/p\mathbb{Z}$.)

Step 3 (powers of 4). If $c \equiv 1 \pmod 3$ is a unit with $c \equiv 4^e \pmod{M_K}$ (Lemma 3.4), then $\phi(c) = 4^e$ in $\mathbb{Z}/p\mathbb{Z}$. This is Step 2 with $j = 2e$.

Step 4 (the self-target and $4^{3^{K-1}} = 1$). Write $K = n + 1$ and $s = 3^n$, so $M_K = 9s$ and $3^K = 3s$. Since $9s$ is odd there is y with $2y + 1 = 9s$. Then $y < 9s$, $y \equiv 1 \pmod 3$ and $y \equiv 4 \pmod 9$. Consider $e_1 = y + 9s$ and $e_2 = y + 18s$. Both are $< 27s = 3^{K+2}$ and $\equiv 4 \pmod 9$, so both are undo sources. Their targets are

$$\tfrac{e_1-1}{3} = y, \qquad \tfrac{e_2-1}{3} = y + 3s = y + 3^K,$$

and both targets are units below M_K. (For $K = 1$: $e = 13 \mapsto 4$ and $e = 22 \mapsto 7$ at $M = 9...$ with $y = 4$, $s = 1$.) By (6.1) and collapse:

$$3\phi(y) = \phi(y) - 1 \;\Rightarrow\; \phi(y) = -\tfrac{1}{2}, \qquad 3\phi(y + 3s) = \phi(y) - 1 = 3\phi(y) \;\Rightarrow\; \phi(y + 3s) = \phi(y).$$

Now write $y \equiv 2^j \pmod{M_K}$ (Lemma 3.2(3)). Using Lemma 3.3, $4^s \equiv 1 + 3s \pmod{9s}$, so

$$2^{j+2s} = 2^j 4^s \equiv y(1 + 3s) = y + 3sy \equiv y + 3s \pmod{9s},$$

because $y = 3m + 1$ gives $3sy = 9sm + 3s$. By Step 2, $\phi(y) = 2^j$ and $\phi(y + 3s) = 2^{j+2s}$. Hence $2^{j+2s} = 2^j$ in $\mathbb{Z}/p\mathbb{Z}$; cancelling $2^j \neq 0$ gives $2^{2s} = 1$, i.e.

$$4^{3^{K-1}} = 1 \quad \text{in } \mathbb{Z}/p\mathbb{Z}. \tag{6.2}$$

Step 5 (the order of 4 in $\mathbb{Z}/p\mathbb{Z}$). By (6.2), $\mathrm{ord}_p(4)$ divides 3^{K-1}, so $\mathrm{ord}_p(4) = 3^a$ with $a \leq K - 1$. If $a = 0$ then $4 = 1$ in $\mathbb{Z}/p\mathbb{Z}$, so $3 = 0$, impossible for $p \geq 5$. Thus $1 \leq a \leq K - 1$. In particular $K = 1$ *is already impossible*, and no further argument is needed in that case. For $K \geq 2$ we have, for all m,

$$4^m = 1 \text{ in } \mathbb{Z}/p\mathbb{Z} \iff 3^a \mid m. \tag{6.3}$$

Step 6 (the coupled nodes). Write $a = a' + 1$ and $r = 3^{a'}$, so $3^a = 3r$ and $3^{a+1} = 9r \leq 3^K$. Let $c = 4 + 9r = 4 + 3^{a+1}$. Then $c < M_K$ (as $4 + 3^K < 3 \cdot 3^K$), and $c \equiv 4 \pmod 9$, so c is an undo source with target $\frac{c-1}{3} = 1 + 3r = 1 + 3^a$.

Value at c. Write $c \equiv 4^e \pmod{M_K}$ with $e = e' + 1$ (note $e \geq 1$ since $c \neq 1$). Reduce modulo $9r$, which divides M_K: $4^{e'+1} \equiv 4 + 9r \equiv 4 \pmod{9r}$. Cancelling 4 gives $4^{e'} \equiv 1 \pmod{3^{a+1}}$, so by Lemma 3.1 (with $K = a$) $3^a \mid e'$, and by (6.3) $4^{e'} = 1$ in $\mathbb{Z}/p\mathbb{Z}$. By Step 3, $\phi(c) = 4^{e'+1} = 4$.

The target. The undo relation (6.1) at c gives $3\Phi(1 + 3r, 1) = \phi(c) - 1 = 3$, so $\Phi(1 + 3r, 1) = 1$ and, by collapse, $\phi(1 + 3r) = 1$.

Contradiction. The unit $1 + 3r \equiv 1 \pmod 3$ is $\equiv 4^d \pmod{M_K}$ for some d (Lemma 3.4), and $\phi(1 + 3r) = 4^d$ by Step 3. So $4^d = 1$ in $\mathbb{Z}/p\mathbb{Z}$, hence $3^a \mid d$ by (6.3), hence $4^d \equiv 1 \pmod{3^{a+1}}$ by Lemma 3.1. But $4^d \equiv 1 + 3r \pmod{9r}$, and $0 < 3r < 9r$, so $1 + 3r \not\equiv 1 \pmod{9r}$. Contradiction. □

Remark 6.3 (Why this is the same obstruction)**.** Step 2 says that ϕ restricted to the units agrees with the character $2^j \mapsto 2^j$. Step 4 shows that this character must send $1 + 3^K$ to 1, and Step 6 shows that it would then have to be trivial on $1 + 3^a$ as well, which has too large an order. The argument avoids introducing the group homomorphism and discrete-logarithm valuations explicitly: only the two facts of Lemma 3.1 and Lemma 3.2 are used.

7. From the obstruction to a translation

Lemma 7.1 (A common fixed point gives a flat assignment)**.** *Let $K \geq 1$. If $x \in \mathbb{Z}/p\mathbb{Z}$ satisfies $h(x) = x$ for all $h \in \mathrm{Hol}(K)$, then a flat assignment exists at depth K.*

Proof. By Lemma 5.3, for each state u choose a walk from base to u with label g_u, and put $\Phi(u) = g_u(x)$. This does not depend on the choice: two walks g_u, g'_u give the closed walk $g_u \,;\, g'^{-1}_u \in \mathrm{Hol}(K)$ which fixes x. If $u \to w$ is an edge with label f then $g_u \,;\, f$ is a walk from base to w, so $\Phi(w) = f(g_u(x)) = f(\Phi(u))$. □

Proposition 7.2 (A translation in the holonomy)**.** *Let $p \geq 5$ and $K \geq 1$. Then there is $d \neq 0$ with $(1, d) \in \mathrm{Hol}(K)$.*

Proof. Suppose not. By Theorem 6.2 and Lemma 7.1, $\mathrm{Hol}(K)$ has no common fixed point. We derive a contradiction by exhibiting one.

Case 1: all slopes are 1. Then every element is $(1, m)$ and, by assumption, has $m = 0$; so $\mathrm{Hol}(K)$ is trivial and 0 is a common fixed point.

Case 2: some $h_0 = (\ell_0, m_0) \in \mathrm{Hol}(K)$ has $\ell_0 \neq 1$. Let $x_0 = m_0/(1 - \ell_0)$, the fixed point of h_0. Take any $k = (\ell_k, m_k) \in \mathrm{Hol}(K)$. The commutator $k \,;\, h_0 \,;\, k^{-1} \,;\, h_0^{-1} \in \mathrm{Hol}(K)$ evaluates to

$$\Big(1,\ \frac{(\ell_0 - 1)(\ell_k x_0 + m_k - x_0)}{\ell_k \ell_0}\Big).$$

By assumption this translation is trivial. Since $\ell_0 \neq 1$, we get $\ell_k x_0 + m_k = x_0$, i.e. $k(x_0) = x_0$. So x_0 is a common fixed point.

Both cases contradict the absence of a fixed point. □

8. Density and the main theorem for $p \geq 5$

Theorem 8.1 (Simultaneous density)**.** *Let $p \geq 5$, v even with $3 \nmid v$, $J \geq 1$, $c < 3^{J+1}$ a unit and $r \in \mathbb{Z}/p\mathbb{Z}$. Then there is $n \in \mathcal{T}(v)$ with $n \equiv c \pmod{3^{J+1}}$ and $(n : \mathbb{Z}/p\mathbb{Z}) = r$.*

Proof. Let K_0 be as in Proposition 4.5 and $K = \max(K_0, J, 1)$. Take $d \neq 0$ with $(1, d) \in \mathrm{Hol}(K)$ (Proposition 7.2). By Proposition 5.2, $t \in F_K(1, 0) \Rightarrow t + d \in F_K(1, 0)$. The fibre is non-empty (Lemma 4.4), so it contains $t_0 + kd$ for all k, which exhaust $\mathbb{Z}/p\mathbb{Z}$ since $d \neq 0$. So $F_K(1, 0) = \mathbb{Z}/p\mathbb{Z}$ and $\sigma_K = p$. By Lemma 4.2(3), $F_K(c, 0) = \mathbb{Z}/p\mathbb{Z}$ for every unit $c < 3^{K+1}$; in particular for the given c. By projection, $r \in F_J(c, 0)$. □

Proposition 8.2 (Case $p \geq 5$ of Theorem 1.1)**.** *Theorem 1.1 holds for $p \geq 5$.*

Proof. Let $v = 2w$; it is even, live, and $v \rightsquigarrow w$. Let $J = B + 1$ and $c = 9 \cdot 3^B - 1$. Then $3 \nmid c$, $c < 3^{J+1}$, and $c > B$. Theorem 8.1 gives $n \in \mathcal{T}(v)$ with $n \equiv c \pmod{3^{J+1}}$ and $(n : \mathbb{Z}/p\mathbb{Z}) = r$. Since $c > 0$, $n \geq c > B$. By Lemma 2.2(2), $n \rightsquigarrow v$, hence $n \rightsquigarrow w$. □

9. Small primes

Proposition 9.1 ($p = 2$)**.** *Theorem 1.1 holds for $p = 2$.*

Proof. For $r = 0$ take $n = 2^{B+1}w$. For $r = 1$, let $t = 3B + 3$ and choose $k \in \{2t, 2t + 1\}$ with $2^k w \equiv 1 \pmod 3$ ($k = 2t$ if $w \equiv 1$, $k = 2t + 1$ if $w \equiv 2 \pmod 3$). Put $n = (2^k w - 1)/3$. Then $3n + 1 = 2^k w$ is even, so n is odd, $\mathcal{C}(n) = 2^k w$, and $n \rightsquigarrow w$; also $n > B$ because $2^k w > 6B + 6$. □

Proposition 9.2 ($p = 3$)**.** *Theorem 1.1 holds for $p = 3$.*

Proof. Choose $k \geq B$ with $2^k w \equiv 1 \pmod 3$ (possible with $k \in \{2B, 2B + 1\}$ as above). For $r = 1$ take $n = 2^k w$, for $r = 2$ take $n = 2^{k+1} w$; both exceed B and reach w by halvings. For $r = 0$: there is $j < 6$ with $2^j w \equiv 1 \pmod 9$; set $k = j + 6(B + 1)$. As $64 \equiv 1 \pmod 9$, $2^k w \equiv 1 \pmod 9$. Put $n = (2^k w - 1)/3$: it is odd, $\mathcal{C}(n) = 2^k w$, so $n \rightsquigarrow w$, $n \equiv 0 \pmod 3$ and $n > B$. □

Proof of Theorem 1.1. Combine Proposition 8.2 with the two propositions above. □

10. Application to cycle equations

A hypothetical cycle with k odd steps and S total steps gives $(2^S - 3^k)n = N$ for a positive integer N depending on the step sequence. A *cycle specification* is a triple $(k, S, N) \in \mathbb{N}^3$.

Theorem 10.1. *Let p be any prime, (k, S, N) a cycle specification with $(2^S - 3^k : \mathbb{Z}/p\mathbb{Z}) \neq 0$, and $v > 0$ with $3 \nmid v$. Then there is n with $n \rightsquigarrow v$ and $(2^S - 3^k : \mathbb{Z}/p\mathbb{Z}) \cdot (n : \mathbb{Z}/p\mathbb{Z}) = (N : \mathbb{Z}/p\mathbb{Z})$.*

Proof. Apply Theorem 1.1 with $B = 0$ to the residue $r = (2^S - 3^k : \mathbb{Z}/p\mathbb{Z})^{-1}(N : \mathbb{Z}/p\mathbb{Z})$. □

So congruence conditions modulo a single prime cannot rule out predecessors satisfying a cycle equation.

11. Relation to the coordinate-based proof

The companion paper *Every Remainder Everywhere* (coordinate version) proves the same theorem. The two arguments share the same architecture: finite fibres over 3-adic cylinders, stabilization of their size, exact action of edge labels on stabilized fibres, a translation obtained as a group commutator, and density from translations. The differences are as follows.

- *Moves.* Here only the two atomic backward steps appear; there is no coordinate $x = (3n + 1)/2$, no multiplier $\mu(x)$ modulo 27, no table of branches R and L, and no first-valid-class stopping rule. Parity is an extra bit in the state.
- *Base point.* The base state is $(1, 0)$, not 8, and the seed is $2w$, not a member of a seed table.
- *Obstruction.* In the coordinate proof a non-trivial dilation $(\lambda, 0) \in \mathrm{Hol}(K)$ is constructed by three cases on $\mathrm{ord}_p(4)$, using pairs of left edges with equal multipliers. Here we prove instead that no flat assignment exists (Theorem 6.2); a translation then follows by contraposition and a commutator (Proposition 7.2). The two arguments are of the same nature: the key in each is that 4 has order 3^a modulo p with $a \leq K - 1$, which clashes with the 3-adic order of $1 + 3^a$.
- *Stabilization.* In both, σ_K is non-increasing and positive. Here the equality of odd and even fibres and the equality of fibres at successive depths is obtained from $|F_K(c, 1)| = \sigma_{K+1}$ (Lemma 4.4(2)).

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Statement	Declaration
Collatz map, reachability	`collatz`, `Reaches`
Lemma 1.2, Corollary 1.3	`collatz_double`, `collatz_pow_mul`, `reaches_trans`, `collatz_eq_iff`, `pred_of_three_dvd`
Atomic tree $\mathcal{T}(v)$	`Tree`
Lemma 2.2	`Tree_live`, `Tree_reaches`, `tree_odd_inv`
Lemma 3.1	`lte_four`, `four_pow_modEq_one_iff`
Lemma 3.2	`two_pow_modEq_one_iff`, `two_pow_inj`, `two_transitive`
Lemma 3.3	`four_pow_form`, `four_pow_cong`
Lemma 3.4	`flat_four_form` (first part)
Fibres, Lemma 4.2	`fibre`, `fibre_proj`, `fibre_D_image`, `fibre_D_ncard_le`, `fibre_card_iter`, `fibre_card_const`, `fibre_D_eq`, `fibre_flag1_sub`, `fibre_U_iff`
Lemma 4.4	`sig`, `sig_antitone`, `sig_pos`, `fibre1_card`
Proposition 4.5	`Stab`, `stabilization`

Statement	Declaration
Affine labels	`Aff`, `aapp`, `athen`, `ainv`, `athen_app`, `athen_assoc`, `ainv_app`, `ainv_athen`
Edges, walks, holonomy	`Edge`, `Walk`, `base`, `Hol`, `Walk.append`, `Walk.symm`, `Hol_mul`, `Hol_inv`
Proposition 5.2	`edge_image`, `walk_image`
Lemma 5.3	`walk_two_pow`, `connected_even`, `connected`
Definition of flatness	`Flat`
Theorem 6.2, Step 1	`flat_collapse`, `flat_double`, `flat_one`
Step 2	`flat_hom`, `two_pow_N`
Step 3, relation (6.1)	`flat_four_form`, `flat_U`
Step 4	`flat_four_pow`
Steps 5–6	`flat_contra`
Theorem 6.2	`no_flat`
Lemma 7.1	`flat_of_fixed`
Proposition 7.2	`translation_of_no_fixed`, `translation_exists`
Theorem 8.1	`main_theorem_core`
Proposition 8.2	`main_theorem_p_ge_5`
Small primes	`main_theorem_p_eq_2`, `main_theorem_p_eq_3`
Theorem 1.1	`main_theorem_infinite`, `infinite_predecessors`
Theorem 10.1	`CycleSpec`, `reaching_predecessor_dense_mod_p`, `cycle_local_obstruction_barrier`
Group-theoretic alternative to Steps 5–6 (not used)	`pow3_form`, `principal_order`